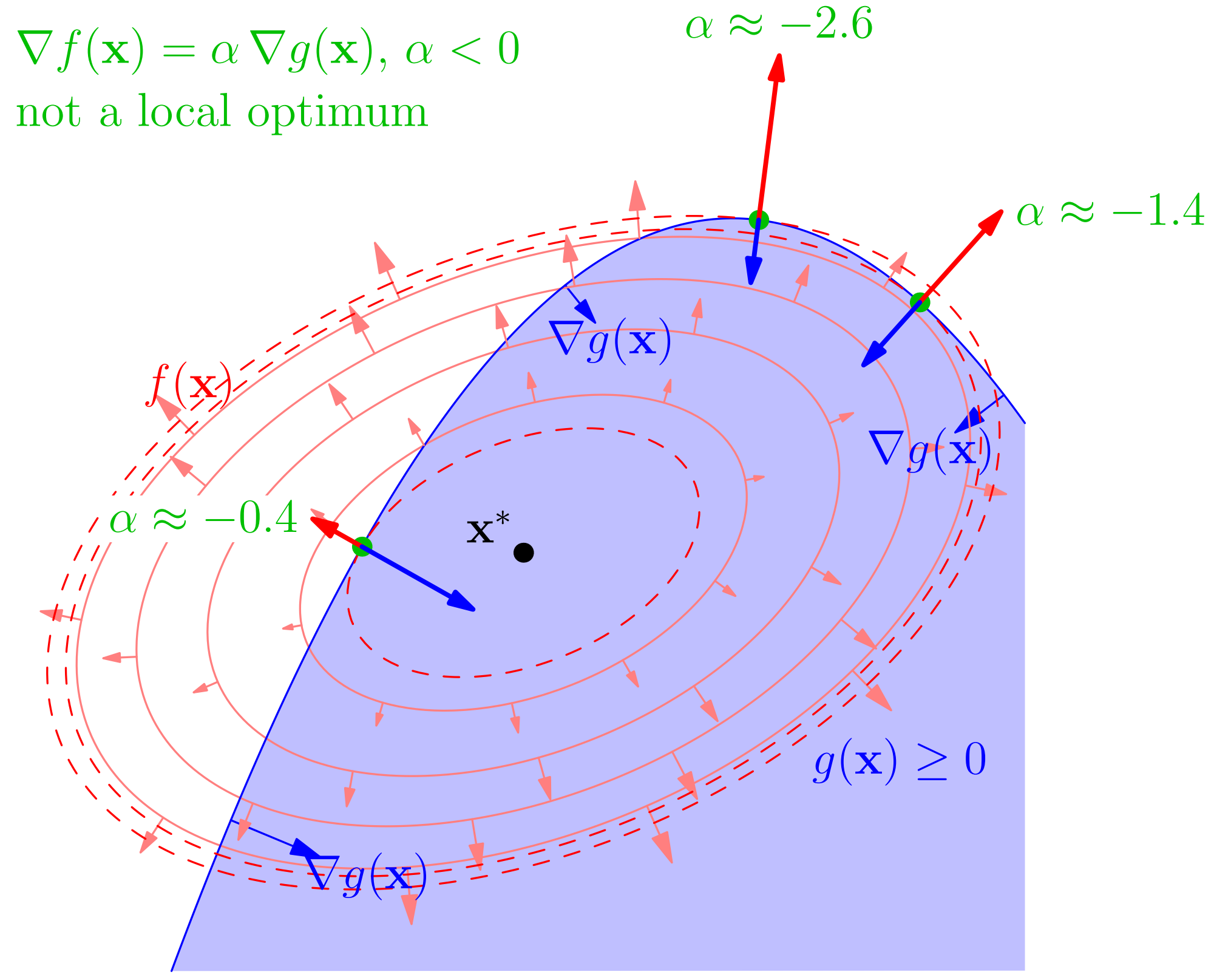
$\nabla f(\mathbf{x}) = \alpha \nabla g(\mathbf{x})$, $\alpha < 0$
not a local optimum
$\alpha \approx -2.6$
$\alpha \approx -1.4$
$\nabla g(\mathbf{x})$
$f(\mathbf{x})$
$\nabla g(\mathbf{x})$
$\alpha \approx -0.4$
$\mathbf{x}^*$
$g(\mathbf{x}) \geq 0$
$\nabla g(\mathbf{x})$